

S&S QUIZ-7 SOLUTION

Sol (1)÷ A causal & stable LTI system S has frequency response,

$$H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega}$$

(a) Given, $H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega}$

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{4+j\omega}{6-\omega^2+5j\omega}$$

$$6 \cdot Y(j\omega) - \omega^2 \cdot Y(j\omega) + 5j\omega \cdot Y(j\omega) = 4X(j\omega) + j\omega \cdot X(j\omega)$$

$$6 \cdot Y(j\omega) + (j\omega)^2 \cdot Y(j\omega) + 5j\omega \cdot Y(j\omega) = 4X(j\omega) + j\omega \cdot X(j\omega) \quad \text{--- (1)}$$

As we know that-

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} \cdot d\omega \quad \text{--- (2)}$$

where, $x(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} X(j\omega)$

Differentiating both side w.r.t. 't', we get -

$$\begin{aligned} \frac{d}{dt} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot (j\omega) \cdot e^{j\omega t} \cdot d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) \cdot X(j\omega) \cdot e^{j\omega t} \cdot d\omega \end{aligned}$$

which shows,

$$\frac{d}{dt} x(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} (j\omega) X(j\omega) \quad \text{--- (3)}$$

Similarly,

$$\frac{d^2}{dt^2} x(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} (j\omega)^2 \cdot X(j\omega) \quad \text{--- (4)}$$

→ (1 Mark)

Taking inverse fourier transform (IFT) of both side of eqn(1) with the help of eqn(3) & eqn(4), we get -

$$6 y(t) + \frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) = 4 x(t) + \frac{d}{dt} x(t)$$

$$\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6 y(t) = \frac{d}{dt} x(t) + 4 x(t)$$

→ (2 Marks)

(b) Given- $H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega} = \frac{4+j\omega}{(2+j\omega)(3+j\omega)}$

$$= \frac{2}{2+j\omega} - \frac{1}{3+j\omega} \quad \text{--- (5)}$$

As we know that -

$$e^{-at} u(t) \xleftrightarrow[\text{IFT}]{\text{FT}} \frac{1}{a+j\omega}, \quad a > 0 \quad \text{--- (6)}$$

Taking inverse fourier transform (IFT) of both side of eqn (5) with the help of eqn(6), we get -

$$h(t) = 2 e^{-2t} u(t) - e^{-3t} u(t) \quad \text{--- (7)} \rightarrow (2 \text{ Marks})$$

(c) Given, $x(t) = e^{-4t} u(t) - t^2 e^{-4t} u(t)$ --- (8)

As we know that -

$$t \cdot e^{-at} u(t) \xleftrightarrow[\text{IFT}]{\text{FT}} \frac{1}{(a+j\omega)^2} \quad \text{--- (7)}$$

$$t^2 \cdot e^{-at} u(t) \xleftrightarrow[\text{IFT}]{\text{FT}} \frac{2}{(a+j\omega)^3} \quad \text{--- (8)}$$

Taking FT of both side of eqⁿ(9) with the help of eqⁿ(6) & eqⁿ(8), we get —

$$X(j\omega) = \frac{1}{4+j\omega} - \frac{2}{(4+j\omega)^3} = \frac{(4+j\omega)^2 - 2}{(4+j\omega)^3} \rightarrow (1 \text{ Marks})$$

Therefore, $Y(j\omega) = H(j\omega) \cdot X(j\omega)$

$$\begin{aligned} &= \frac{(4+j\omega)}{(2+j\omega)(3+j\omega)} \times \frac{(4+j\omega)^2 - 2}{(4+j\omega)^3} \\ &= \frac{(4+j\omega)^2 - 2}{(2+j\omega)(3+j\omega)(4+j\omega)^2} \\ &= \frac{1/2}{2+j\omega} + \frac{1}{3+j\omega} - \frac{3/2}{4+j\omega} - \frac{1}{(4+j\omega)^2} \end{aligned} \quad \text{--- (10)}$$

Taking IFT of both side of eqⁿ(10) with the help of eqⁿ(6) & eqⁿ(7), we get —

$$y(t) = \frac{1}{2} e^{-2t} u(t) + e^{-3t} u(t) - \frac{3}{2} e^{-4t} u(t) - t e^{-4t} u(t) \rightarrow (2 \text{ Marks})$$

Proof:

(I) $x(t) = e^{-at} u(t)$, $a > 0$

$$\begin{aligned} \therefore X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(a+j\omega)t} dt = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty} \\ &= \frac{-1}{(a+j\omega)} [e^{-(a+j\omega)\infty} - e^0] = \frac{-1}{(a+j\omega)} (0 - 1) \\ &= \left(\frac{1}{a+j\omega} \right) \rightarrow (1 \text{ Marks}) \end{aligned}$$

$$(II) \quad x(t) = t e^{-at} u(t), \quad a > 0$$

As we know that — $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot (-jt) dt$$

$$j \frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} [t x(t)] e^{-j\omega t} dt$$

which shows —

$$t x(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} j \frac{d}{d\omega} X(j\omega)$$

Similarly, $t^n x(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} (j)^n \frac{d^n}{d\omega^n} X(j\omega)$

Hence, $t e^{-at} u(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} \frac{1}{(a+j\omega)^2}$

$$t^2 e^{-at} u(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} \frac{2}{(a+j\omega)^3}$$

⋮

$$t^n e^{-at} u(t) \xrightleftharpoons[\text{IFT}]{\text{FT}} \frac{n!}{(a+j\omega)^{n+1}} \rightarrow (1 \text{ Marks})$$